

TDI Python Data Analysis Capstone Project

Project Title: Real-World Data Analysis Challenge: Industry-Specific Insights

Dear Python Data Analyst,

Congratulations on reaching the final week of the TDI Python Data Analysis programme. You have put in the work, and this capstone is your opportunity to show what you are truly capable of.

This project is entirely yours. You will choose an industry, find your own data, ask the right questions, and deliver insights that matter. There are no step-by-step instructions from here. This is you, your Python skills, and a real dataset.

We look forward to seeing your analytical thinking and creativity come to life.

Scenario Overview

You are an individual data analyst tackling a high-stakes project within your chosen industry. Your organisation depends on the insights you uncover to make data-driven decisions, innovate, and maintain a competitive edge. This project challenges you to identify key trends, patterns, and actionable insights that can guide strategic initiatives, optimize operations, and drive success.

Your Mission

Your task is to conduct an in-depth analysis of data within an industry of your choice. Your findings will be presented to senior management, who will use your insights to shape future strategies. This is a unique opportunity to apply your Python skills in a real-world scenario, leveraging data to deliver impactful insights.

Project Objectives

1. Data Collection and Connection

- Find relevant data within your chosen industry by downloading datasets from reputable sources or by connecting directly to websites or APIs.
- Ensure the data you gather is accurate, comprehensive, and reflects the current state of the industry.

- Reputable sources include Kaggle, Google Dataset Search, Maven Analytics, UCI Machine Learning Repository, World Bank Open Data, or any other credible platform of your choosing.
- Do not use AI-generated data.

2. Tool Utilization

- Python is your primary tool for data cleaning, transformation, analysis, and visualization.
- You may also use additional tools like SQL or Excel if they complement your workflow, but Python should be the primary platform for analysis and visualization.

3. Data Cleaning and Analysis

- Begin by cleaning the data to ensure accuracy. This involves removing inconsistencies, duplicates, and handling missing values.
- Conduct a thorough analysis to uncover trends, correlations, and patterns within the data, utilizing both quantitative and qualitative methods.

4. Developing Research Questions

- Formulate at least three clear and actionable research questions that will guide your analysis.
- These questions should be tailored to address specific challenges and opportunities within your selected industry.

5. Optional: Prediction or Forecasting

- If your data supports it, you may include a simple prediction or forecast as part of your analysis.
- This could be a trend forecast, a simple linear regression, or a basic classification. Keep it relevant to your research questions.

Video Resources

The following videos may help you as you work through the project:

#	Topic	Link	Focus
1	End-to-End Data Science Projects	Ken Jee: Portfolio Project From Scratch	Project structure and pipeline
2	Building ROI and Business Cases	Storytelling with Data: Crash Course	Translating analysis into business value

#	Topic	Link	Focus
3	Data Visualization for Executives	Storytelling with Data #36: Presenting Data	Clear, compelling charts and dashboards
4	Technical Documentation	How To Write A README File (2025)	READMEs, notebooks, reproducibility
5	Presenting Data Analysis Results	Storytelling with Data: Mini-Workshop	Storytelling with data
6	Portfolio Project Development	Ken Jee: 5 Impactful Portfolio Projects	GitHub, showcasing work, job applications

Final Deliverables

1. Python Notebook

- Your primary deliverable is a well-documented Jupyter Notebook containing your code, analysis process, and visualizations.

2. PowerPoint Presentation

- Create a concise presentation for senior management, showcasing your findings with a narrative supported by data and visuals.

3. Word Document (or PDF)

- Submit a detailed report that includes your research questions, methodology, data sources, analysis process, and conclusions.
- This document should also cover the tools used, challenges faced, and how they were overcome.

4. Additional Files (if applicable)

- Include any SQL scripts, Python scripts, or datasets used in the project.

Timeline

You have 14 days to complete this project. Use this time effectively to explore your data, refine your analysis, and craft compelling insights.

Evaluation Criteria

Your work will be evaluated based on the following criteria:

Data Sourcing and Relevance: The quality and appropriateness of your chosen data.

Data Cleaning and Preparation: The thoroughness and accuracy of your data cleaning process.

Analysis and Insights: The depth, accuracy, and relevance of the analysis conducted.

Visualization: The clarity, accuracy, and effectiveness of your visualizations in communicating insights.

Documentation and Presentation: The professionalism, clarity, and structure of your report, presentation, and Python notebook.

Submission

A Google Form will be shared at the end of the 14-day project period. Submit all your deliverables through that form. Look out for it in the WhatsApp group.

This project immerses you in a real-world data analysis context where your work has the potential to drive significant decisions within your industry. It encourages you to think critically, work independently, and apply your skills in a strategic, impactful way. The success of this project depends on your ability to deliver clear, data-driven insights that can influence your organisation's direction.

Good luck and enjoy your data-driven journey!

Sincerely,
The Projects Team,
The Data Immersed